

Offset Experiment

In the assumption, our assumed distribution for the j -th site is:

$$f_j(y_{ij}; \theta_j) = \lambda_j f(y_{ij}; \mu_1) + (1 - \lambda_j) f(y_{ij}; \mu_0)$$

Offset test

We assume that there is a mean shift in the data at the j -th site:

$$\begin{aligned} Y_{ij} &\sim \lambda_j \mathcal{N}(\mu_1 + \delta_{j1}, \Sigma) + (1 - \lambda_j) \mathcal{N}(\mu_0 + \delta_{j0}, \Sigma) \\ \mu_0 &= 4, \mu_1 = 5, \Sigma = \sigma^2 I_d, \\ \delta_{jk} &\sim \mathcal{N}(0, \tau^2 I_d). \end{aligned}$$

We test four methods:

Distributed EM, Pooled EM, Average Estimator, Shift-Distributed EM

Among them, Shift-Distributed EM is the improved method.

The experimental data are set as follows:

Parameter	Value
The sample size for each site n	1000
site number K	10
dimension d	5
Covariance σ^2	2.5
mixture imbalance a	0.1
τ	$\{0, 0.1, 0.25, 0.5, 1.0\}$

Metrics:

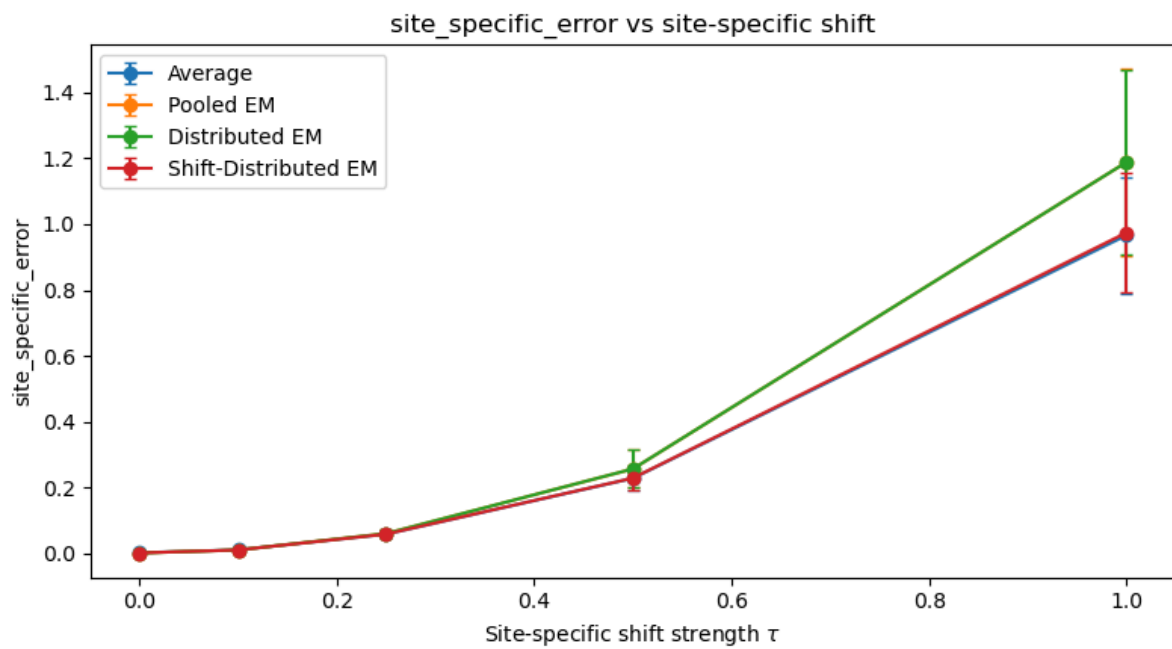
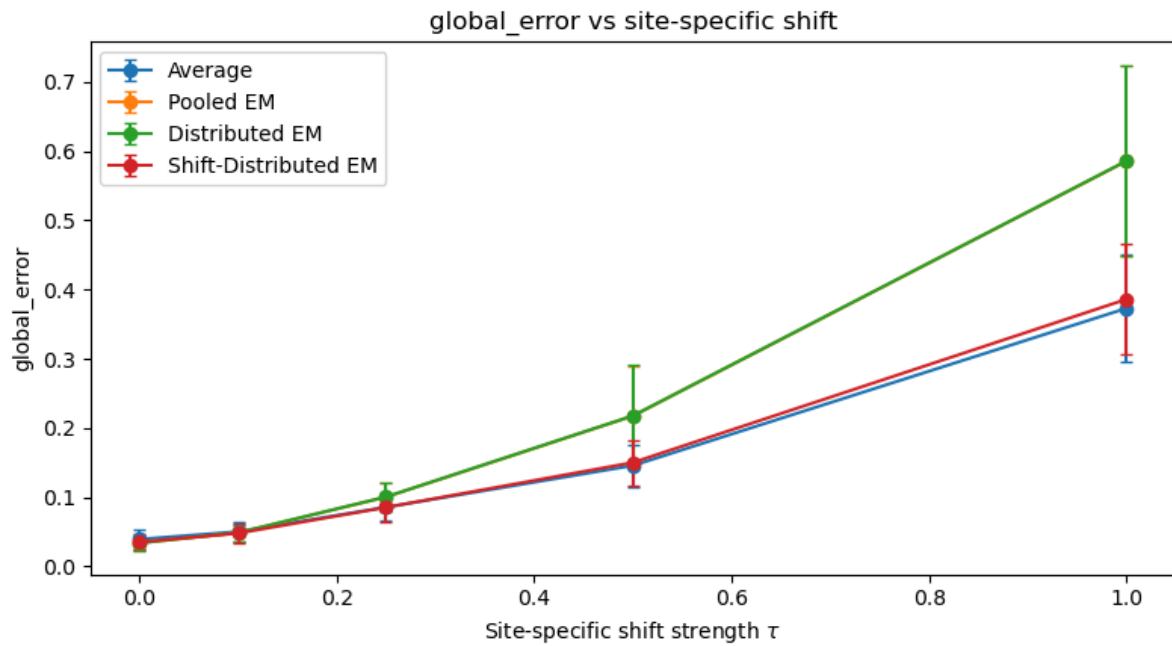
1. Global Error

$$\text{GlobalError} = \frac{\sqrt{\|\hat{\mu}_0 - \mu_0\|^2 + \|\hat{\mu}_1 - \mu_1\|^2}}{\sqrt{2d}}$$

2. Site Error

$$\text{SiteError} = \frac{1}{K} \sum_{j=1}^K (\|\hat{\mu}_0 - (\mu_0 + \delta_{j0})\|^2 + \|\hat{\mu}_1 - (\mu_1 + \delta_{j1})\|^2)$$

Results:



(Note: The Pooled EM method overlaps with the Distributed EM method too much, making it difficult to distinguish clearly.)

The experimental results show that the estimation error of all methods increases as the site-specific shift strength τ grows. Among them, the original Distributed EM and Pooled EM are the most sensitive to distributional shift, with errors increasing rapidly under large heterogeneity. In contrast, the proposed Shift-Distributed EM significantly reduces the estimation error and maintains more stable performance under stronger site-specific shifts, demonstrating improved robustness to cross-site distributional heterogeneity.